

CURVED PROJECTIVE LEVELS FORCE MAXIMAL NORM GROWTH

AUTOMATED PROOF ATTEMPT FOR ARXIV:2308.10582

ABSTRACT. We strengthen the stationary-phase criterion in the source paper: a nondegenerate extremum is unnecessary. Maximal L^p multiplier-norm growth already follows if one regular level hypersurface of the homogeneous phase has nonzero projective Gauss–Kronecker curvature at one point. This gives a complete negative answer to the source question in dimension three. Using a compact-developable classification theorem of Ishikawa–Morimoto, it also gives a negative answer for even phases in dimensions four and six. Thus any affirmative example must have every regular projective level developable; the existence of such a global phase, and the required analytic upper bound, remain open in the unrestricted higher-dimensional case.

1. SOURCE QUESTION AND EXACT SCOPE

Let $\Phi \in C^\infty(\mathbb{R}^d \setminus \{0\}, \mathbb{R})$ be homogeneous of degree zero, and let T_Φ^t be the Fourier multiplier with symbol $e^{it\Phi}$. Closing Remark 1 of Bulj [1] asks exactly:

“When $d \geq 3$, does there exist a nonconstant unimodular homogeneous function $\Phi \in C^\infty(\mathbb{R}^d \setminus \{0\}, \mathbb{R})$ of degree 0 such that

$$\liminf_{t \rightarrow \infty} \frac{\|T_\Phi^t\|_{L^p \rightarrow L^p}}{t^{d|\frac{1}{2} - \frac{1}{p}|}} = 0?$$

”

The exact source context is reproduced below.

4. CLOSING REMARKS

Remark 1. The following question remains open and it could be interesting.

When $d \geq 3$, does there exist a nonconstant unimodular homogeneous function $\Phi \in C^\infty(\mathbb{R}^d \setminus \{0\}, \mathbb{R})$ of degree 0 such that

$$\liminf_{t \rightarrow \infty} \frac{\|T_\Phi^t\|_{L^p \rightarrow L^p}}{t^{d|\frac{1}{2} - \frac{1}{p}|}} = 0?$$

Remark 2. When $d = 2$, one can modify the approach from the proof of [I Theorem 3] to prove the sharp estimate both in p and t for all nonconstant 0-homogeneous unimodular functions $\Phi \in C^\infty(S^1, \mathbb{R})$, but it is not clear how to extend that approach to $d \geq 3$ nor it is apparent how to modify the current approach to imply the sharpness of the estimate both in p and t .

The question is understood for a fixed $p \in (1, \infty)$. At $p = 2$ the operator is unitary and the denominator has exponent zero, so the answer is trivially no. The results below apply simultaneously to every such p .

2. PROOF INTUITION

The source changes variables from frequency ξ to an affine direction $u \in \mathbb{R}^{d-1}$ and a radial coordinate r . At a stationary point, the Hessian of the transformed phase is not $H\phi$ but the bordered matrix

$$\begin{pmatrix} H\phi & -\nabla\phi \\ -\nabla\phi^\top & 0 \end{pmatrix}.$$

At a regular point this matrix is invertible exactly when the Hessian of ϕ restricted to the tangent space of its level set is nondegenerate. That is precisely nonzero Gauss–Kronecker curvature of the projective

level. The source's implicit-function and stationary-phase proof then works locally without any critical point of ϕ .

In dimension three, a regular level curve failing this criterion everywhere must be a great circle. Two different regular values would therefore give two disjoint great circles, which is impossible. For even phases in dimensions four and six, the same intersection argument uses the theorem that compact developable hypersurfaces in $\mathbb{RP}^3$ and $\mathbb{RP}^5$ are projective hyperplanes.

3. THE BORDERED-HESSIAN CRITERION

Write a central affine chart as

$$\phi(u) = \Phi(u, 1), \quad u \in \mathbb{R}^{d-1}.$$

At a regular point u_0 , put $a = \nabla\phi(u_0) \neq 0$ and $E = a^\perp \subset \mathbb{R}^{d-1}$.

Lemma 1 (Bordered determinant). *The matrix*

$$B(u_0) = \begin{pmatrix} H\phi(u_0) & -a \\ -a^\top & 0 \end{pmatrix}$$

is invertible if and only if the quadratic form $H\phi(u_0)|_E$ is nondegenerate.

Proof. Choose an orthonormal basis whose first $d-2$ vectors span E and whose last vector is $a/|a|$. In that basis, for some matrix C , vector b , and scalar h , the bordered matrix is

$$\begin{pmatrix} C & b & 0 \\ b^\top & h & -|a| \\ 0 & -|a| & 0 \end{pmatrix}.$$

Expanding in the last row and then the last column shows that its determinant is nonzero exactly when $\det C \neq 0$. The matrix C represents $H\phi(u_0)|_E$. $\square$

For the regular hypersurface $\{\phi = \phi(u_0)\}$, the second fundamental form in this affine chart is, up to the nonzero factor $|a|^{-1}$, $H\phi(u_0)|_E$. Thus Lemma 1 is equivalently the existence of nonzero Gauss–Kronecker curvature. Nondegeneracy of the projective second fundamental form is invariant under changing central charts.

Theorem 2 (Curved regular level). *Let $d \geq 2$, $p \in (1, \infty)$, and let Φ be a smooth real nonconstant degree-zero homogeneous phase. Suppose that at some regular point of $\Phi|_{S^{d-1}}$, a regular projective level hypersurface has nondegenerate second fundamental form. Then there is $c = c(d, p, \Phi) > 0$ such that*

$$\|T_\Phi^t\|_{L^p \rightarrow L^p} \geq ct^{d|\frac{1}{2} - \frac{1}{p}|} \quad (t \geq 1).$$

In particular, the normalized liminf in the source question is positive.

Proof. Rotate coordinates and choose a central chart containing the given point. With u_0 and a as above, consider the transformed phase used in the source paper,

$$\Psi_x(u, r) = \phi(u) + r(u \cdot x_- + x_d).$$

Set

$$r_0 = 1, \quad x_-^0 = -a, \quad x_d^0 = u_0 \cdot a.$$

Then $(u_0, 1)$ is a stationary point of Ψ_{x^0} , since

$$\nabla_u \Psi_{x^0}(u_0, 1) = a + x_-^0 = 0, \quad \partial_r \Psi_{x^0}(u_0, 1) = u_0 \cdot x_-^0 + x_d^0 = 0.$$

Its full (u, r) Hessian is

$$H_{(u,r)} \Psi_{x^0}(u_0, 1) = \begin{pmatrix} H\phi(u_0) & -a \\ -a^\top & 0 \end{pmatrix},$$

which is invertible by Lemma 1.

Apply the implicit-function theorem to $F((u, r), x) = \nabla_{(u,r)} \Psi_x(u, r)$. After shrinking neighbourhoods, there are open sets $U \ni x^0$ and $V \ni (u_0, 1)$ and a unique smooth map $g : U \rightarrow V$ such that $g(x)$ is the

unique stationary point in V , and its Hessian stays invertible. Choose the Fourier transform of a fixed Schwartz test function to be supported in the image of a sufficiently small compact subset of V under the source's diffeomorphism $(u, r) \mapsto (ru, r)$ and to equal one on a smaller subset.

The proof of Bulj's main theorem now applies verbatim from this point. Uniform stationary phase gives

$$|T_{\Phi}^t f(tx)| \geq c_0 t^{-d/2}$$

on a fixed positive-measure open subset of U , for all sufficiently large t . Scaling tx therefore yields, when $1 < p \leq 2$,

$$\|T_{\Phi}^t f\|_p \geq c_1 t^{d/p-d/2}.$$

Division by the fixed $\|f\|_p$ gives the operator lower bound. The case $p \geq 2$ follows from duality and $(T_{\Phi}^t)^* = T_{\Phi}^{-t}$, exactly as in the source. Enlarging constants over a compact initial t -interval gives the stated form for $t \geq 1$. $\square$

Corollary 3 (Necessary geometry of any counterexample). *If a phase satisfies the zero normalized $\liminf$ asked for in the source, then every connected component of every regular spherical level is locally projectively developable: in every central chart its projective Gauss map has rank strictly below $d - 2$ everywhere.*

Proof. Otherwise Theorem 2 applies at a point of full Gauss rank. $\square$

4. A COMPLETE ANSWER IN DIMENSION THREE

Theorem 4 (No submaximal phase in $d = 3$). *For every nonconstant smooth real degree-zero homogeneous phase on $\mathbb{R}^3 \setminus \{0\}$ and every $p \in (1, \infty)$,*

$$\liminf_{t \rightarrow \infty} \frac{\|T_{\Phi}^t\|_{L^p \rightarrow L^p}}{t^{3|\frac{1}{2} - \frac{1}{p}|}} > 0.$$

Consequently the source question has a negative answer in dimension three.

Proof. Suppose the curvature condition in Theorem 2 fails at every regular point of $\Phi|_{S^2}$. A regular level component is a compact smooth curve. In every central projective chart its second fundamental form vanishes, so it is locally a projective line. Lifting back to S^2 , it is locally an arc of a great circle.

On a connected component these local great circles agree on overlaps: a great circle is determined by any nontrivial open arc. Hence the component lies in one great circle. It is both open in that great circle (by the local description) and closed (by compactness), so it is the whole great circle.

The range of the nonconstant function $\Phi|_{S^2}$ is a nondegenerate compact interval. Sard's theorem supplies two distinct regular values c_1, c_2 in its interior. Choose one component of each nonempty level. The preceding paragraph makes them two great circles. Any two great circles on S^2 intersect, while distinct level sets are disjoint, a contradiction. Therefore a curved regular point exists, and Theorem 2 finishes the proof. $\square$

5. EVEN PHASES IN DIMENSIONS FOUR AND SIX

We use the following published projective-geometric fact, attributed there to Ishikawa–Morimoto.

Theorem 5 (Ishikawa–Morimoto [2]). *Every compact developable C^∞ hypersurface in $\mathbb{RP}^3$ or $\mathbb{RP}^5$ is a projective hyperplane.*

The same source explains why no unrestricted all-dimensional replacement is available: nonsingular compact cubic developables exist in $\mathbb{RP}^4, \mathbb{RP}^7, \mathbb{RP}^{13}, \mathbb{RP}^{25}$.

Theorem 6 (Even phases). *Let $d \in \{4, 6\}$ and suppose in addition that $\Phi(-\xi) = \Phi(\xi)$. If Φ is nonconstant, then for every $p \in (1, \infty)$,*

$$\liminf_{t \rightarrow \infty} \frac{\|T_{\Phi}^t\|_{L^p \rightarrow L^p}}{t^{d|\frac{1}{2} - \frac{1}{p}|}} > 0.$$

Proof. Evenness makes $\Phi|_{S^{d-1}}$ descend through the antipodal quotient to a smooth nonconstant function $\bar{\Phi}$ on $\mathbb{RP}^{d-1}$. Suppose no regular level has a curved point. Choose two distinct regular values c_1, c_2 in the interior of the range, and a connected component M_i of each level. Each M_i is a compact embedded smooth hypersurface in $\mathbb{RP}^{d-1}$ and is developable everywhere. Since $d-1$ is 3 or 5, Theorem 5 says that both M_i are projective hyperplanes. Two projective hyperplanes in the same real projective space always intersect, contradicting $M_1 \subset \bar{\Phi}^{-1}(c_1)$ and $M_2 \subset \bar{\Phi}^{-1}(c_2)$. Apply Theorem 2. $\square$

6. A BROAD CANDIDATE FAMILY THAT CANNOT WORK

Corollary 7 (All nonconstant quadratic Rayleigh phases). *Let A be a real symmetric nonscalar $d \times d$ matrix and*

$$\Phi_A(\xi) = \frac{\xi^\top A \xi}{|\xi|^2}.$$

For every $d \geq 2$ and $p \in (1, \infty)$, Φ_A has maximal norm growth in the sense of Theorem 2.

Proof. Choose c strictly between the smallest and largest eigenvalues of A and not itself an eigenvalue. The projective level $\Phi_A = c$ is the nonempty quadric

$$\xi^\top (A - cI) \xi = 0.$$

The bilinear form $Q = A - cI$ is invertible and indefinite. At a null vector ξ , the tangent hyperplane is $\{v : Q(\xi, v) = 0\}$. If a tangent vector v lies in the radical of the induced second fundamental form, then $Q(v, \cdot)$ annihilates that tangent hyperplane, hence is proportional to $Q(\xi, \cdot)$. Invertibility gives $v \in \mathbb{R}\xi$, which is zero in the projective tangent space. Thus the projective second fundamental form is nondegenerate, and Theorem 2 applies. $\square$

7. LIMITATIONS, NOVELTY, AND HUMAN REVIEW

The unrestricted source question remains open here for $d \geq 4$. The precise remaining geometric obstruction is now sharp: an affirmative example must make every regular level developable. Compact nontrivial developables show that this condition cannot be dismissed pointwise. One must either rule out a global foliation by such levels or build one and then prove a multiplier upper bound. Degenerate stationary phase alone is not an upper bound and can increase oscillatory contributions.

No priority claim is made. The bordered-Hessian calculation and the dimension-three argument appear elementary but should be checked by a harmonic analyst and a projective geometer. The even $d = 4, 6$ result depends on applying Theorem 5 to embedded regular-level components after descent to projective space; this is why the evenness hypothesis is retained. The literature search found no later paper claiming these consequences of the source method.

REFERENCES

- [1] A. Bulj, *Generic norm growth of powers of homogeneous unimodular Fourier multipliers*, arXiv:2308.10582, 2023.
- [2] G. Ishikawa, *Developable hypersurfaces and homogeneous spaces in a real projective space*, Lobachevskii J. Math. **3** (1999), 113–125; Hokkaido University Preprint Series in Mathematics 449, doi:10.14943/83595.